

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 2

Year: 2019

Subject: 394172 Mathematics II

Section: 15-18

Date: 5 March 2020

Time: 13.00-16.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 10 questions. There will be 10 pages (including this page) and they are worth 100 points.

- Answer the questions on this test paper.
- This is a closed book examination.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Question 1. 1.1 Find solutions of the following nonlinear equations (5 points)

$$x^2 - 2xy + y^2 = 3,$$

$$x^2 + xy + y^2 = 12.$$

1.2 Solve the following nonlinear equations (5 points)

$$x^2 + y^2 = 25,$$

$$|x| + y = 5.$$

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Question 2. 2.1 If (x, y, z) is a solution of the following system of equations

$$x + y + z = 1,$$

$$x + y - 2z = 3,$$

$$2x + y + z = 2,$$

Then find the value of xyz . (5 points)

2.2 Evaluate the value of $\sqrt{9 - 3\sqrt{8}}$. (5 points)

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Question 3. 3.1 Determine whether the following statements are True or False.

a) If a is a real number and $a \neq 0$ then $y = \left(\frac{a^2}{a^2 + 1} \right)^x$ is a decreasing function.

(1 point)

b) The equation $2^x = 21$ has no solution. (2 points)

c) If $x = a$ is a solution of the equation $\left(\frac{1}{2} \right)^x + \left(\frac{2}{3} \right)^x = 1$ then $a > 1$. (2 points)

3.2 Let A be a set of all solutions for the equation

$$(4^x + 2^x - 6)^2 = (4^x - 2)^2 + (2^x - 4)^2$$

Find the sum of all members in A . (5 points)

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Question 4. 4.1 If $\frac{2}{\sqrt{12-2\sqrt{35}}} + \frac{3}{\sqrt{7-2\sqrt{10}}} - \frac{5}{\sqrt{9-2\sqrt{14}}} = a\sqrt{b}$, where a and b are positive numbers, then find the value of $b-a$. (5 points)

4.2 Solve the equation $x^2 + 6\sqrt{x^2 - 2x + 5} = 11 + 2x$. (5 points)

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Question 5. Given $A = \log \sqrt[3]{(9^{-1})(27)^{-\frac{4}{3}}}$ and $B = \log \frac{25}{8} - 2 \log \frac{5}{3} + \log \frac{24}{9}$.

Evaluate the expression $\frac{A}{B}$. (10 points)

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Question 6. Find the solutions of the inequality

$$\log_{\frac{1}{4}} \log_{\frac{1}{3}} \log_{\frac{1}{2}} \sqrt[3]{\frac{1}{x^2 - 3x + 4}} \leq 0. \quad (10 \text{ points})$$

Question 7. Given the system of equations

$$\begin{aligned} 9^{\log_3 a} + 4^{\log_2 b} &= 24, \\ \log_3 a - \log_{\frac{1}{3}} b &= 2 - \log_3 2. \end{aligned}$$

What is the value of $|a^2 - b^2|$? (10 marks)

Question 8. Let a, x , and y be integers.

8.1 Determine whether the statement “If $a^3 \mid 4x^2 + y$ and $a^2 \mid 3x^2 - 3y$, then $a \mid 15x^2$ ” is correct or not. Give an example to support your answer. (5 points)

8.2 Find all positive integer n which satisfies $(8n + 9) \mid (12n + 5)$. (5 points)

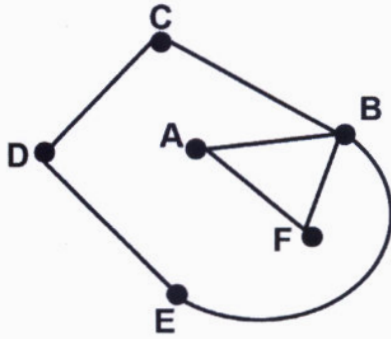
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Question 9. 9.1 Find $\gcd(192, 276)$ using Euclidean algorithm and then find $\text{lcm}(192, 276)$. (5 points)

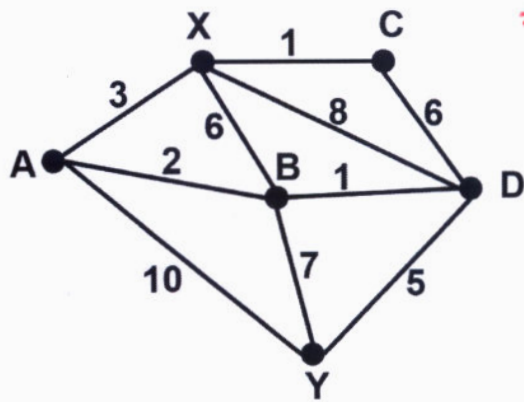
9.2 If $\gcd(192, 276) = 192x + 276y$, then find $x + y$. (5 points)

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Question 10. 10.1 Determine whether the following graph has an Euler circuit. If it has, find an Euler circuit of this graph. (5 points)



10.2 Find the shortest path between the vertex X and the vertex Y in the weighted graph. (5 points)



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